

B.TECH · 1ST YEAR · 1ST SEMESTER

BEEE Revision Sheet

Basic Electrical & Electronics Engineering

UNIT I — Semiconductor Devices

Diodes · Transistors · Amplifiers · Characteristics

LAST-DAY REVISION WEAPON | EXAM-READY FORMAT | HIGH-RETENTION

- ▶ Introduction to Electronics & Evolution
- ▶ Semiconductor Basics & Energy Bands
- ▶ PN Junction Diode
- ▶ Zener Diode & Voltage Regulation
- ▶ Bipolar Junction Transistor (BJT)
- ▶ Transistor Configurations (CB / CE / CC)
- ▶ CE Amplifier (Small Signal)
- ▶ Formulas, Flashcards & Key Terms
- ▶ Quick Revision Sheet & Exam Tips

⚡ DEFINITION

Electronics is the branch of science and engineering that deals with the behavior of electrons in vacuum, gases, and semiconductors, and with the utilization of devices based on these phenomena.

🌐 Importance in Modern Engineering

- ▶ Foundation of communication, computation, and control systems
- ▶ Enables miniaturization — from mainframes to smartphones
- ▶ Core of embedded systems, IoT, robotics, and smart devices
- ▶ Essential in power electronics, signal processing, and automation
- ▶ Drives medical devices (ECG, MRI) and aerospace systems

📅 Evolution of Electronics — Timeline

Era	Technology	Key Features	Limitations
1904–1950s Vacuum Tube Era	Vacuum Tubes (Diodes, Triodes)	Amplification, radio communication, early computers (ENIAC)	Bulky, high power consumption, short lifespan, generates heat
1947–1960s Transistor Era	Bipolar Junction Transistors (BJT)	Compact, low power, reliable; portable radios & early computers	Individual components; manual circuit assembly required
1958–present IC Era	Integrated Circuits (SSI → VLSI)	Millions of components on chip; high speed, low cost	Complex fabrication; heat dissipation challenges
2000s–present Nano Era	Nano-electronics (FinFET, 2D materials)	Sub-10 nm features; extreme miniaturization, quantum effects	Quantum tunneling, leakage currents, fabrication cost

📌 MEMORY HOOK

V → T → IC → N · Vacuum → Transistor → Integrated Circuit → Nano

🌀 DEFINITION

A semiconductor is a material whose electrical conductivity lies between that of a conductor and an insulator ($\sigma \approx 10^{-4}$ to 10^4 S/m). Examples: Silicon (Si), Germanium (Ge), GaAs.

ENERGY BAND CONCEPT

- ▶ **Valence Band:** Highest filled energy band at 0 K
- ▶ **Conduction Band:** Next higher empty band
- ▶ **Band Gap (E_g):** Si = 1.1 eV, Ge = 0.67 eV, GaAs = 1.43 eV
- ▶ Electrons jump valence→conduction with thermal/light energy

🔌 Charge Carriers

ELECTRONS

- ▶ Negatively charged free carriers
- ▶ Move in conduction band
- ▶ Majority carriers in N-type
- ▶ Minority carriers in P-type
- ▶ Move opposite to conventional current

HOLES

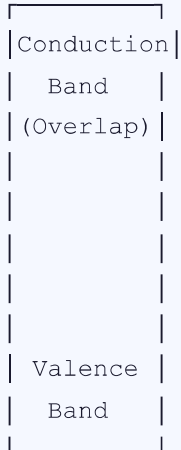
- ▶ Positively charged virtual carriers
- ▶ Absence of electron in valence band
- ▶ Majority carriers in P-type
- ▶ Minority carriers in N-type
- ▶ Move in direction of conventional current

📊 Semiconductor Types Comparison

Property	Intrinsic Semiconductor	N-type (Extrinsic)	P-type (Extrinsic)
Definition	Pure semiconductor; no impurities	Doped with pentavalent impurity	Doped with trivalent impurity
Dopant	None	Phosphorus (P), Arsenic (As), Antimony (Sb)	Boron (B), Aluminium (Al), Indium (In)
Majority Carrier	Electrons = Holes	Electrons (e^-)	Holes (h^+)
Minority Carrier	Electrons = Holes	Holes (h^+)	Electrons (e^-)
Conductivity	Very low (increases with temp)	High (controlled by doping)	High (controlled by doping)
Fermi Level	Middle of band gap	Near conduction band	Near valence band
Net Charge	Neutral	Neutral (ionized donors fixed)	Neutral (ionized acceptors fixed)

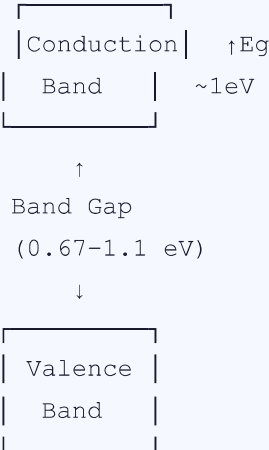
ENERGY BAND DIAGRAM COMPARISON

CONDUCTOR



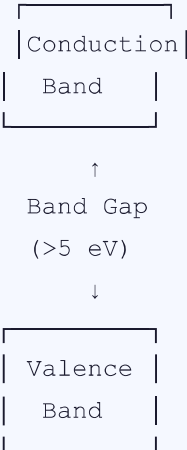
σ very high

SEMICONDUCTOR



σ moderate

INSULATOR



σ very low

▲ DEFINITION

A PN junction diode is a two-terminal semiconductor device formed by joining P-type and N-type semiconductor materials, creating a one-way valve for current flow.

Formation & Depletion Region

- ▶ When P-type and N-type are joined, holes from P diffuse to N; electrons from N diffuse to P
- ▶ This creates a **Depletion Region** — a zone depleted of free carriers at the junction
- ▶ Fixed ionized charges set up an internal electric field (**built-in electric field**)
- ▶ **Barrier Potential (V_0):** Si = 0.7 V, Ge = 0.3 V
- ▶ Diffusion stops when drift current = diffusion current (equilibrium)

STRUCTURE OF PN JUNCTION (Equilibrium)

P-REGION	DEPLETION	N-REGION
Holes (h^+) (mobile) (+ + + + +)	[- - - + + +] Ionized acceptors & donors (fixed) ← E-field →	Electrons (e^-) (mobile) (- - - - -)
Anode (A)	← Barrier Potential → $V_0 = 0.7V$ (Si)	Cathode (K)

Symbol: A  K (Triangle points toward N-side)

Working Principle

✓ FORWARD BIAS

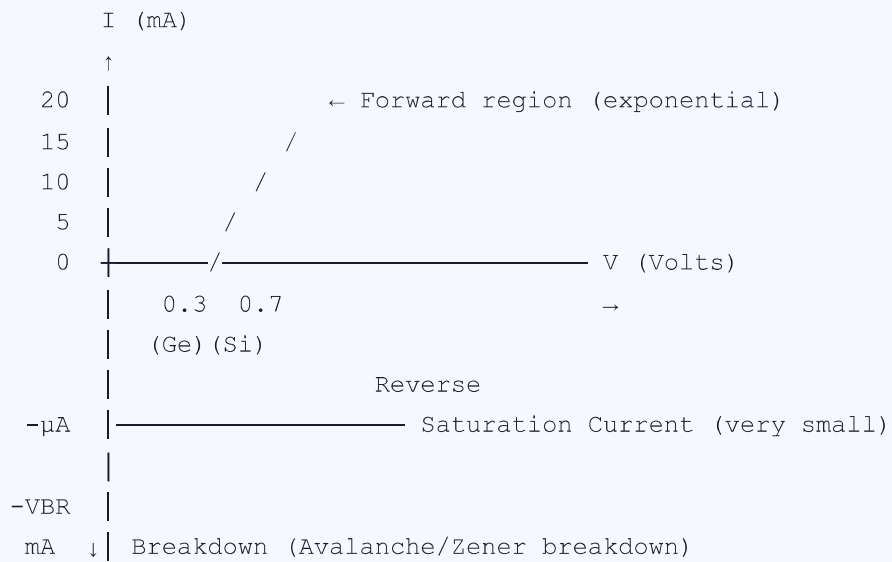
- ▶ +ve terminal of battery → P-side; -ve → N-side
- ▶ Applied voltage opposes barrier potential
- ▶ Depletion region **narrows**
- ▶ Majority carriers cross the junction
- ▶ Current flows easily → LOW resistance
- ▶ Cut-in voltage: Si = 0.7V, Ge = 0.3V

⊖ REVERSE BIAS

- ▶ -ve terminal → P-side; +ve → N-side
- ▶ Applied voltage aids barrier potential
- ▶ Depletion region **widens**
- ▶ Majority carriers repelled from junction
- ▶ Only tiny reverse saturation current (I_0) flows
- ▶ HIGH resistance; used as open switch

V-I Characteristics

V-I CHARACTERISTIC CURVE (PN Diode)



REGIONS:

1. Dead Zone : 0 to cut-in voltage (no significant current)
2. Forward Bias : Above cut-in $\rightarrow$ exponential current rise
3. Reverse Bias : Small reverse saturation current I_o
4. Breakdown : At V_{BR} $\rightarrow$ sudden current surge

📦 Applications

- ✓ Half-wave and full-wave rectifiers
- ✓ Clipping and clamping circuits
- ✓ Demodulation of AM signals
- ✓ Protection circuits (freewheeling diode)
- ✓ Logic gates (diode logic)

⚡ DEFINITION

A Zener diode is a specially doped PN junction diode designed to operate in the reverse-breakdown region, maintaining a constant voltage across it regardless of current variation.

🔬 Zener Effect (Breakdown)

ZENER BREAKDOWN

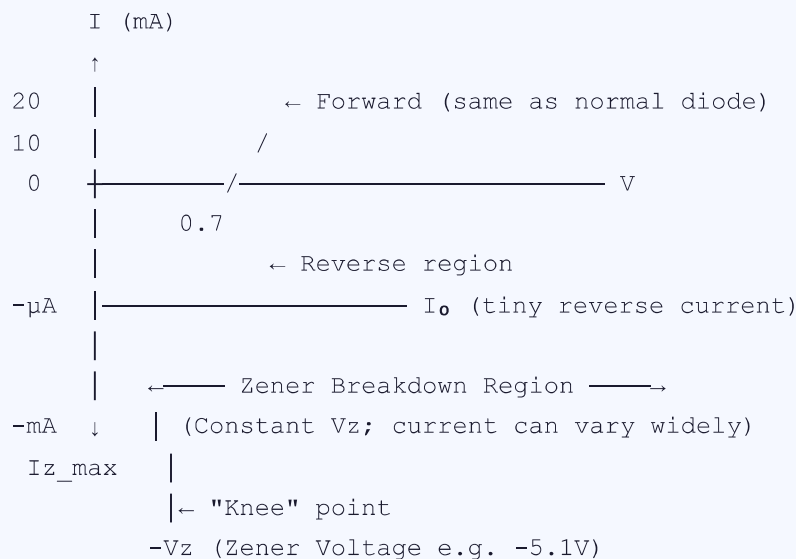
- ▶ Occurs at **low reverse voltages** ($< 5V$)
- ▶ High electric field across thin depletion layer
- ▶ Direct rupture of covalent bonds
- ▶ Electrons pulled out without collision
- ▶ Sharp, well-defined breakdown
- ▶ Has **negative temperature coefficient**

AVALANCHE BREAKDOWN

- ▶ Occurs at **high reverse voltages** ($> 8V$)
- ▶ Minority carriers gain sufficient kinetic energy
- ▶ Collide with lattice atoms $\rightarrow$ ionization
- ▶ Multiplication effect (avalanche)
- ▶ Gradual breakdown
- ▶ Has **positive temperature coefficient**

📈 V-I Characteristics

V-I CHARACTERISTIC CURVE (Zener Diode)

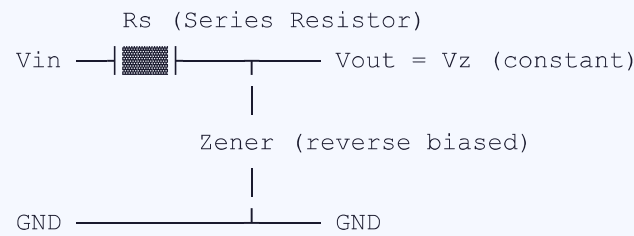


KEY POINT: In the Zener region, V_z remains CONSTANT despite large changes in current I_z .

Symbol: A $\begin{array}{c} | \\ \text{---} \text{Z} \text{---} \\ | \end{array}$ K (bent cathode bar)

🔧 As Voltage Regulator

ZENER VOLTAGE REGULATOR CIRCUIT



Condition: $V_{in} > V_z$ always
Rs drops excess voltage: $V_s = V_{in} - V_z$
Zener current: $I_z = (V_{in} - V_z) / R_s - I_L$

✓ **ADVANTAGES**

- ▶ Simple voltage regulation circuit
- ▶ Low cost, compact
- ▶ Sharp, stable breakdown voltage
- ▶ Available in wide range of voltages (2.4V–200V)

✗ **LIMITATIONS**

- ▶ Power dissipation limits usage
- ▶ Not efficient for large load current variation
- ▶ Output voltage affected by temperature
- ▶ Noisy output compared to IC regulators

 PN Diode vs Zener Diode

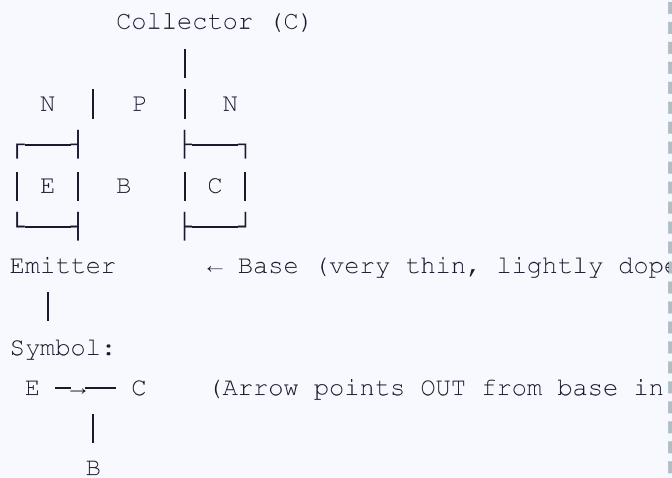
Property	PN Junction Diode	Zener Diode
Operating Region	Forward bias (normally)	Reverse breakdown (normally)
Breakdown Voltage	Destructive; damages diode	Designed to operate here safely
Doping	Moderate doping	Heavily doped (thin depletion layer)
Primary Use	Rectification, switching	Voltage regulation, reference
Symbol	Straight cathode bar (⌋ ⌋)	Bent/Z-shaped cathode bar (⌋ ⌋Z⌋)
Reverse Current	I_o (very small, leakage)	Increases significantly at V_z

DEFINITION

A Bipolar Junction Transistor (BJT) is a three-terminal semiconductor device consisting of two PN junctions, used for amplification and switching. The term "bipolar" refers to the use of both electrons and holes for current conduction.

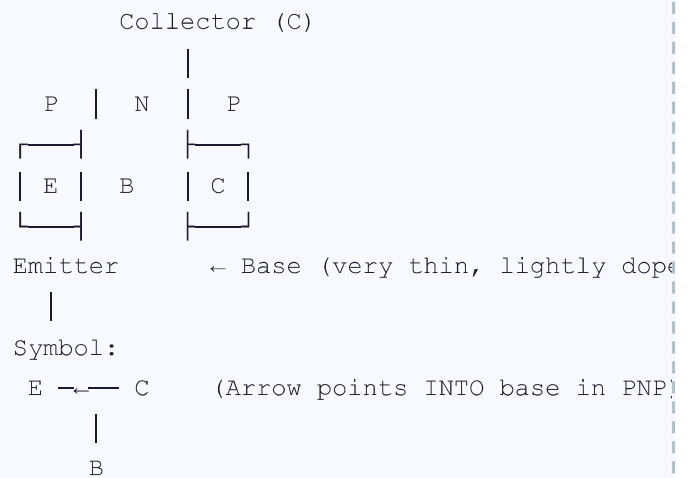
Types & Structures

NPN TRANSISTOR



Majority carriers: Electrons

PNP TRANSISTOR



Majority carriers: Holes

Terminal Functions

Terminal	Doping	Size	Function
Emitter (E)	Heavily doped	Medium	Emits/injects majority carriers into base
Base (B)	Lightly doped	Very thin (~1–10 μm)	Controls carrier flow; most recombination here
Collector (C)	Moderately doped	Largest	Collects carriers from base; dissipates power

Working Principle (NPN)

- ▶ **EB junction:** Forward biased → Emitter injects electrons into base
- ▶ **CB junction:** Reverse biased → Creates strong drift field to collect electrons
- ▶ ~95–99% of injected electrons reach collector (very thin base → low recombination)
- ▶ Small base current I_B controls large collector current I_C
- ▶ Current relations: $I_E = I_B + I_C$

TRANSISTOR ACTION KEY IDEA

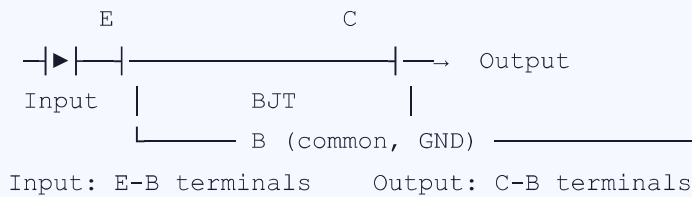
A small signal at the Base controls a much larger current at the Collector — this is the **amplifying action**. Think of it as a controlled valve: small input → large output.

★ INTRODUCTION

A transistor has 3 terminals. One terminal is common to both input and output. Based on which terminal is common, three configurations are possible: **Common Base (CB)**, **Common Emitter (CE)**, **Common Collector (CC)**.

1 Common Base (CB) Configuration

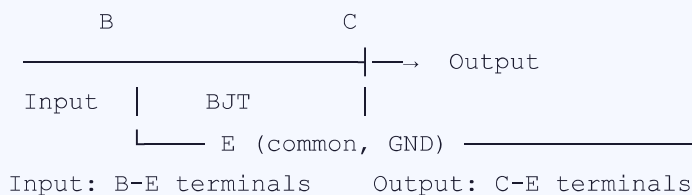
COMMON BASE CIRCUIT (NPN)



- ▶ **Input characteristics:** I_E vs V_{EB} (with V_{CB} constant) — similar to PN diode forward curve
- ▶ **Output characteristics:** I_C vs V_{CB} (with I_E constant) — flat, horizontal lines ($I_C \approx I_E$)
- ▶ **Current gain $\alpha = I_C / I_E$** (0.95 – 0.999; always < 1)
- ▶ High input impedance ❌ (actually LOW — Base is common so input is E-B → low r_{in})
- ▶ Used in: HF amplifiers, impedance matching

2 Common Emitter (CE) Configuration

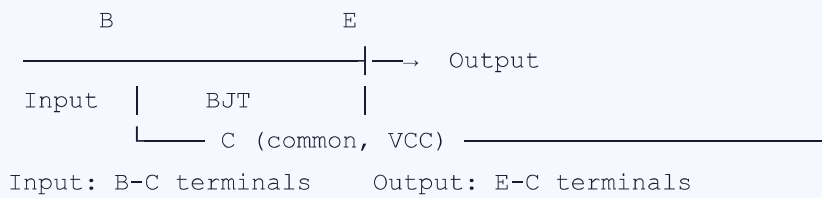
COMMON EMITTER CIRCUIT (NPN)



- ▶ **Input characteristics:** I_B vs V_{BE} (with V_{CE} constant) — PN diode forward curve
- ▶ **Output characteristics:** I_C vs V_{CE} (with I_B constant) — flat after saturation knee
- ▶ **Current gain $\beta = I_C / I_B$** (20–500; $> 1 \rightarrow$ amplification)
- ▶ Moderate input resistance, high output resistance
- ▶ **Phase inversion of 180°** between input and output
- ▶ Most widely used configuration for amplifiers & switching

3 Common Collector (CC) / Emitter Follower

COMMON COLLECTOR CIRCUIT (NPN)



- ▶ **Current gain $\gamma = I_E / I_B = (1 + \beta)$**
- ▶ Voltage gain ≈ 1 (slightly less)
- ▶ Very high input impedance, very low output impedance
- ▶ **No phase inversion** — output follows input
- ▶ Used for: impedance matching, buffer amplifiers

Configuration Comparison Table (MUST MEMORIZE)

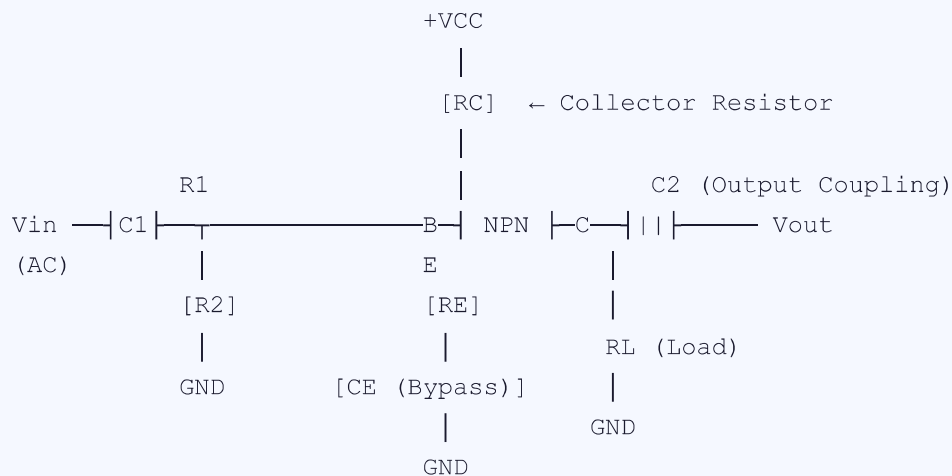
Parameter	Common Base (CB)	Common Emitter (CE)	Common Collector (CC)
Input Terminal	Emitter	Base	Base
Output Terminal	Collector	Collector	Emitter
Common Terminal	Base	Emitter	Collector
Input Resistance	Low ($\sim 30\text{--}150\ \Omega$)	Medium ($\sim 1\text{--}2\ \text{k}\Omega$)	Very High ($\sim 100\ \text{k}\Omega\text{--}1\ \text{M}\Omega$)
Output Resistance	Very High ($\sim 500\ \text{k}\Omega$)	High ($\sim 40\text{--}50\ \text{k}\Omega$)	Very Low ($\sim 50\text{--}1000\ \Omega$)
Current Gain	$\alpha < 1$ ($0.95\text{--}0.999$)	$\beta = 20\text{--}500$	$\gamma = (1 + \beta) > 1$
Voltage Gain	High	Very High	≈ 1 (less than 1)
Power Gain	High	Very High (highest)	High
Phase Shift	0° (no inversion)	180° (inverted)	0° (no inversion)
Applications	RF / HF amplifiers, current buffer	Audio amplifiers, oscillators, switching	Buffer, impedance matching, drivers

 DEFINITION

A Common Emitter (CE) amplifier is a transistor amplifier circuit where the emitter terminal is common to both input and output, widely used for voltage amplification with phase inversion.

 Circuit Diagram (Text-Based)

CE AMPLIFIER CIRCUIT

**COMPONENTS:**

R1, R2 → Voltage divider bias (sets Q-point / DC bias)
 RC → Collector resistor (provides voltage drop)
 RE → Emitter resistor (stability; bypassed by CE for AC)
 CE → Bypass capacitor (shorts RE for AC signals)
 C1, C2 → Coupling capacitors (block DC, pass AC)

 Working Principle

- ▶ **DC Biasing:** R1-R2 divider sets base voltage → establishes Q-point (operating point)
- ▶ **AC Input:** Superimposed on DC bias via coupling capacitor C1
- ▶ **Amplification:** Small variation in I_B → large variation in I_C → large voltage across RC
- ▶ **Output:** Amplified AC voltage taken across RC (or RL) via coupling cap C2
- ▶ **Phase Reversal:** When V_{in} increases (+ve half) → $I_B \uparrow$ → $I_C \uparrow$ → $V_{CE} \downarrow$ → V_{out} is -ve half

 VOLTAGE GAIN FORMULA

- ▶ $A_v = -\beta \times (RC \parallel RL) / r_e$
- ▶ Where $r_e = 26\text{mV} / I_C$ (dynamic emitter resistance)

 KEY POINTS

- ▶ Highest power gain of all configurations
- ▶ Input and output are 180° out of phase

- ▶ Negative sign → 180° phase inversion
- ▶ Typical gain: 100–500

- ▶ Q-point should be at center of load line
- ▶ CE bypass capacitor CE increases gain

Applications

- ✓ Audio pre-amplifiers and power amplifiers
- ✓ Radio frequency (RF) amplifiers
- ✓ Switching circuits (digital logic)
- ✓ Signal conditioning in sensor circuits
- ✓ First stage of multi-stage amplifiers

⚡ DIODE CURRENT EQUATION (SHOCKLEY'S EQUATION)

$$I = I_o [e^{(V/\eta VT)} - 1]$$

where: I_o = Reverse saturation current (μA range)

V = Applied voltage across diode

η = Ideality factor (1 for Ge, 2 for Si)

VT = Thermal voltage = $kT/q \approx 26 \text{ mV}$ at 300 K

k = Boltzmann's constant = $1.38 \times 10^{-23} \text{ J/K}$

q = Electron charge = $1.6 \times 10^{-19} \text{ C}$

📋 BJT CURRENT RELATIONS

$$I_E = I_B + I_C \quad (\text{KCL at transistor})$$

$$\alpha \text{ (Alpha)} = I_C / I_E \quad [\text{Common Base gain, } 0 < \alpha < 1]$$

$$\beta \text{ (Beta)} = I_C / I_B \quad [\text{Common Emitter gain, typically 20-500}]$$

$$\gamma \text{ (Gamma)} = I_E / I_B \quad [\text{Common Collector gain}]$$

Relationships between α and β :

$$\beta = \alpha / (1 - \alpha) \quad \alpha = \beta / (1 + \beta)$$

$$\gamma = 1 + \beta = I_E / I_B$$

📐 CE AMPLIFIER KEY FORMULAS

$$r_e = V_T / I_C = 26\text{mV} / I_C \quad (\text{Dynamic emitter resistance})$$

$$A_v = -\beta \times (R_C \parallel R_L) / r_e \quad (\text{Voltage gain with } R_L)$$

$$A_i = I_C / I_B = \beta \quad (\text{Current gain})$$

$$A_p = A_v \times A_i \quad (\text{Power gain})$$

$$\text{Input Resistance: } R_{in} = R_1 \parallel R_2 \parallel (\beta \times r_e)$$

$$\text{Output Resistance: } R_o \approx R_C$$

⚡ ZENER REGULATOR FORMULAS

$V_{out} = V_z$ (Regulated output voltage)

$R_s = (V_{in} - V_z) / (I_z + I_L)$ (Series resistor)

$I_z = I_s - I_L$ (Zener current)

$P_z = V_z \times I_{z(max)}$ (Max power dissipation)

$I_L = V_z / R_L$ (Load current)

⚡ THERMAL VOLTAGE

$V_T = kT/q$

At $T = 27^\circ\text{C}$ (300 K): $V_T = 25.9 \text{ mV} \approx 26 \text{ mV}$

At $T = 0^\circ\text{C}$ (273 K): $V_T = 23.5 \text{ mV}$

Semiconductor → Material with conductivity between conductor & insulator (Si, Ge)

Extrinsic Semiconductor → Semiconductor doped with impurities to increase conductivity

Donor Impurity → Pentavalent atom (P, As) that donates free electrons in N-type

Hole → Absence of electron in covalent bond; acts as positive charge carrier

Depletion Region → Zone near PN junction depleted of mobile charge carriers

Diffusion Current → Current due to carrier movement from high to low concentration region

Forward Bias → Positive terminal to P-side; reduces barrier → large current flows

Reverse Saturation Current (I_0) → Tiny current in reverse bias due to minority carriers

Breakdown Voltage (VBR) → Reverse voltage at which diode undergoes controlled breakdown

Zener Breakdown → Direct bond rupture by strong electric field in reverse bias (<5V)

Zener Voltage (V_z) → Constant reverse breakdown voltage of a Zener diode

BJT (Bipolar Junction Transistor) → 3-terminal semiconductor device using both electrons & holes

Base → Very thin, lightly doped BJT terminal that controls carrier flow

Alpha (α) → CB current gain = I_C/I_E ; always less than 1 (0.95–0.999)

Gamma (γ) → CC current gain = $I_E/I_B = (1+\beta)$; greater than 1

Load Line → Straight line on output characteristics representing circuit constraints

Cut-off Region → BJT region where both junctions are reverse

Intrinsic Semiconductor → Pure semiconductor with equal electrons and holes

Doping → Process of adding controlled impurities to a pure semiconductor

Acceptor Impurity → Trivalent atom (B, Al) that accepts electrons; creates holes in P-type

PN Junction → Interface formed by joining P-type and N-type semiconductor materials

Barrier Potential → Built-in potential at junction; Si = 0.7V, Ge = 0.3V

Drift Current → Current due to carrier movement under an applied electric field

Reverse Bias → Negative terminal to P-side; increases barrier → tiny current flows

Cut-in Voltage → Minimum forward voltage for significant conduction (0.3V Ge, 0.7V Si)

Avalanche Breakdown → Carrier multiplication by high-energy collision in reverse bias (>8V)

Zener Diode → Heavily doped diode designed for stable operation in reverse breakdown

Voltage Regulator → Circuit that maintains constant output voltage despite input/load variation

Emitter → Heavily doped BJT terminal that injects/emits majority carriers

Collector → Moderately doped, large terminal that collects injected carriers

Beta (β) → CE current gain = I_C/I_B ; typically 20–500; the amplification factor

Q-Point → Quiescent/operating point; DC bias point on transistor characteristics

Saturation Region → BJT region where both junctions are forward biased; transistor fully ON

Active Region → BJT region for amplification (EB forward, CB

biased; transistor fully OFF

reverse biased)

CE Amplifier → Transistor amplifier with emitter common; highest power gain configuration

Phase Inversion → 180° phase shift between input & output in CE amplifier

Coupling Capacitor → Capacitor used to pass AC signals while blocking DC component

Bypass Capacitor → Capacitor across RE that shorts it for AC signals, increasing voltage gain

Voltage Gain (A_v) → Ratio of output voltage to input voltage;
 $A_v = V_{out}/V_{in}$

Thermal Voltage (V_T) → $kT/q \approx 26\text{mV}$ at 300K; appears in diode equation

Q: What is a PN junction?

A: Interface formed by joining P-type and N-type semiconductor materials with a depletion region at the boundary.

Q: What happens at the moment of junction formation?

A: Holes diffuse from P to N, electrons diffuse from N to P, forming a depletion region and built-in electric field.

Q: What is barrier potential for Silicon?

A: 0.7 V (for Germanium it is 0.3 V).

Q: What happens to depletion region in forward bias?

A: It narrows/reduces, allowing majority carrier flow and current to conduct.

Q: What happens to depletion region in reverse bias?

A: It widens/expands, blocking majority carrier flow; only small reverse saturation current flows.

Q: What is reverse saturation current I_0 ?

A: Tiny current in reverse-biased diode due to minority carrier drift; typically in μA (Si) or mA range (Ge).

Q: Write Shockley's diode equation.

A: $I = I_0 [e^{(V/\eta VT)} - 1]$ where $VT = kT/q \approx 26 \text{ mV}$ at 300 K.

Q: What is Zener breakdown?

A: Sudden increase in reverse current due to strong electric field rupturing covalent bonds; occurs below $\sim 5\text{V}$.

Q: Difference between Zener and Avalanche breakdown?

A: Zener: field ionization at low voltage ($<5\text{V}$), negative temp coeff. Avalanche: impact ionization at high voltage ($>8\text{V}$), positive temp coeff.

Q: Why is Zener diode used as voltage regulator?

A: Because Zener voltage (V_z) remains constant despite changes in input voltage or load current.

Q: What does BJT stand for?

A: Bipolar Junction Transistor — "bipolar" because both electrons and holes participate in conduction.

Q: What is the biasing condition for BJT active region?

A: Emitter-Base junction: forward biased; Collector-Base junction: reverse biased.

Q: Write the fundamental BJT current equation.

A: $I_E = I_B + I_C$ (by Kirchhoff's Current Law at the transistor node).

Q: Define alpha (α) of a transistor.

A: $\alpha = I_C / I_E$ = Common Base current gain; always less than 1 (typically 0.95–0.999).

Q: Define beta (β) of a transistor.

A: $\beta = I_C / I_B$ = Common Emitter current gain; typically 20–500; the key amplification parameter.

Q: Relation between α and β ?

A: $\beta = \alpha / (1 - \alpha)$ and $\alpha = \beta / (1 + \beta)$. They are mathematically interdependent.

Q: Why is Base region made very thin?

A: To minimize recombination of injected carriers so that $\sim 95\text{--}99\%$ reach the Collector, ensuring high gain.

Q: Which BJT configuration gives highest power gain?

A: Common Emitter (CE) configuration has the highest power gain among all three configurations.

Q: Which configuration is used as voltage buffer?

A: Common Collector (CC) / Emitter Follower — voltage gain ≈ 1 , very high input resistance.

Q: What is phase inversion in CE amplifier?

A: Output signal is 180° out of phase with input — when input goes +ve, output goes -ve due to transistor action.

Q: What is the purpose of coupling capacitors C1, C2?

A: They block DC component and allow only AC signals to pass between amplifier stages or to the load.

Q: What is bypass capacitor CE in CE amplifier?

A: It shorts the emitter resistor RE for AC signals, preventing AC signal degeneration and increasing voltage gain.

Q: What is Q-point (operating point)?

A: The quiescent DC bias point of the transistor; ideally at center of AC load line for maximum swing.

Q: What is the saturation region of BJT?

A: Both EB and CB junctions forward biased; transistor is fully ON; $V_{CE} \approx 0.2V$ (Si).

Q: What is cut-off region of BJT?

A: Both junctions reverse biased; $I_B \approx 0$; $I_C \approx 0$; transistor is fully OFF.

Q: What is dynamic emitter resistance r_e ?

A: $r_e = V_T/I_C = 26mV/I_C$; small AC resistance of emitter junction used in gain calculations.

Q: What is voltage gain of CE amplifier?

A: $A_v = -\beta(R_C \parallel R_L)/r_e$; negative sign indicates 180° phase reversal.

Q: What is thermal voltage V_T ?

A: $V_T = kT/q \approx 26$ mV at 300 K (room temperature); appears in diode and transistor equations.

Q: Name the majority carriers in N-type semiconductor.

A: Electrons (e^-) are majority carriers; holes are minority carriers in N-type material.

Q: Name the majority carriers in P-type semiconductor.

A: Holes (h^+) are majority carriers; electrons are minority carriers in P-type material.

Q: What is the band gap of Silicon?

A: $E_g(\text{Si}) = 1.1$ eV. For Germanium: $E_g(\text{Ge}) = 0.67$ eV. For GaAs: 1.43 eV.

Q: What is ideality factor η in diode equation?

A: $\eta = 1$ for Germanium (ideal), $\eta = 2$ for Silicon; accounts for recombination in depletion region.

Q: What are the four regions of diode V-I curve?

A: Dead zone, Forward conduction region, Reverse saturation region, and Breakdown region.

Q: What is Common Base current gain γ (gamma)?

A: $\gamma = I_E/I_B = (1 + \beta)$; used for Common Collector configuration gain calculation.

Q: Which configuration has no phase inversion?

A: Common Base (CB) and Common Collector (CC) — both have 0° phase shift between input and output.

⚠ Common Mistakes to Avoid

- ▶ Confusing **Zener** (low V, field ionization) with **Avalanche** (high V, impact)
- ▶ Mixing up **CB, CE, CC** common terminals and their gains
- ▶ Writing $\alpha > 1$ — alpha is always < 1 !
- ▶ Forgetting that CE has 180° phase inversion; CB and CC do not
- ▶ Ignoring the ideality factor η in Shockley's equation
- ▶ Confusing majority/minority carriers across P and N types
- ▶ Stating that depletion region widens in forward bias (it narrows!)
- ▶ Neglecting to draw characteristics curves when asked
- ▶ Not labeling axes (I vs V) on V-I characteristic diagrams
- ▶ Mixing Ge and Si cut-in voltages (0.3V vs 0.7V)

✓ WRITING TIPS FOR FULL MARKS

- ✓ Always draw neat V-I characteristic curves — examiners love diagrams!
- ✓ Label all axes, regions, and key points on curves
- ✓ Use comparison tables for CB vs CE vs CC — guaranteed marks
- ✓ State the formula before substituting numerical values
- ✓ Write definitions precisely — avoid vague language
- ✓ When drawing transistor circuits, label VCC, RC, RB clearly
- ✓ For Zener regulator: draw the circuit, then derive the formula
- ✓ Start answers with definition → working principle → characteristics → applications
- ✓ Highlight key values: V_Z , $V_{BE} = 0.7V$, $V_T = 26mV$
- ✓ Check: $I_E = I_B + I_C$ — use this to verify numerical answers

** FREQUENTLY ASKED QUESTIONS**

- ▶ Explain PN junction formation and depletion region
- ▶ V-I characteristics of PN diode & Zener diode
- ▶ Compare CB, CE, CC configurations (table form)
- ▶ Working of CE amplifier with circuit diagram
- ▶ Explain voltage regulation using Zener diode
- ▶ Derive relation between α and β
- ▶ Shockley's diode equation — explain each term

** MARKS STRATEGY (16-MARK QUESTIONS)**

- ▶ **2 marks:** Definition (1–2 sentences)
- ▶ **4 marks:** Working principle (clear, concise)
- ▶ **4 marks:** Circuit / Structural diagram
- ▶ **4 marks:** Characteristics curve (neat + labeled)
- ▶ **2 marks:** Applications (3–4 bullet points)
- ▶ → Total: Clean structure = full marks!

⚙️ SEMICONDUCTORS

- ◆ Si band gap = 1.1 eV
- ◆ Ge band gap = 0.67 eV
- ◆ N-type → electrons (majority)
- ◆ P-type → holes (majority)
- ◆ Intrinsic: $n_i = p_i$
- ◆ Donor: Pentavalent (P, As)
- ◆ Acceptor: Trivalent (B, Al)

▲ PN DIODE FACTS

- ◆ $V_B(\text{Si}) = 0.7\text{V}$, $V_B(\text{Ge}) = 0.3\text{V}$
- ◆ Forward → depletion narrows
- ◆ Reverse → depletion widens
- ◆ $I = I_0[e^{(V/\eta VT)} - 1]$
- ◆ $V_T = 26\text{mV}$ @ 300K
- ◆ $\eta = 1(\text{Ge}), 2(\text{Si})$
- ◆ Reverse: only I_0 flows

⚡ ZENER DIODE

- ◆ Works in reverse breakdown
- ◆ $V_Z = \text{constant}$ (regulates)
- ◆ Zener breakdown < 5V
- ◆ Avalanche breakdown > 8V
- ◆ Zener: -ve temp coeff
- ◆ Avalanche: +ve temp coeff
- ◆ $R_s = (V_{in} - V_Z)/(I_Z + I_L)$

🔧 BJT BASICS

- ◆ $I_E = I_B + I_C$
- ◆ $\alpha = I_C/I_E (<1)$
- ◆ $\beta = I_C/I_B (20-500)$
- ◆ $\gamma = I_E/I_B = 1 + \beta$
- ◆ $\beta = \alpha/(1 - \alpha)$
- ◆ $\alpha = \beta/(1 + \beta)$
- ◆ Active: EB fwd, CB rev

📊 CONFIGURATIONS

- ◆ CB: $\alpha < 1$, no inversion
- ◆ CE: $\beta > 1$, 180° inversion
- ◆ CC: $\gamma = 1 + \beta$, no inversion
- ◆ CE: highest power gain
- ◆ CC: highest R_{in}
- ◆ CB: lowest R_{in}
- ◆ CC: lowest R_{out}

📈 CE AMPLIFIER

- ◆ $A_v = -\beta(R_C || R_L)/r_e$
- ◆ $r_e = 26\text{mV}/I_C$
- ◆ Phase reversal = 180°
- ◆ CE bypass cap → ↑gain
- ◆ C_1, C_2 = coupling caps
- ◆ R_1, R_2 = voltage divider
- ◆ Q-pt = center of load line

🚀 LAST-MINUTE KEYWORDS

Depletion Region Barrier Potential Zener Voltage Cut-in Voltage Alpha (α) Beta (β) Phase Inversion
 Thermal Voltage Q-Point Active Region Saturation Cut-off Voltage Gain Rectification Load Line
 $I_E = I_B + I_C$ Forward Bias Reverse Bias Shockley Equation Voltage Regulator Emitter Follower Band Gap
 Avalanche Majority Carrier

⚡ CRITICAL NUMBER FACTS

- ▶ Si barrier potential = **0.7 V**
- ▶ Ge barrier potential = **0.3 V**
- ▶ Si band gap = **1.1 eV**
- ▶ Ge band gap = **0.67 eV**
- ▶ Thermal voltage V_T = **26 mV @ 300K**
- ▶ $\eta(\text{Ge}) = 1$, $\eta(\text{Si}) = 2$
- ▶ α typical value = **0.95–0.999**
- ▶ β typical value = **20–500**
- ▶ CE phase shift = **180°**
- ▶ CB, CC phase shift = **0°**
- ▶ $V_{CE(\text{sat})}$ for Si NPN $\approx$ **0.2 V**
- ▶ $V_{BE(\text{active})}$ for Si NPN $\approx$ **0.7 V**